

Ultra-high adsorption capacity of porous calcium carbonate for removal of malachite green from water

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Abstract: A template carbonization strategy was employed to synthesize porous calcium carbonate (CaCO₃) for malachite green (MG) removal. The porous CaCO₃ exhibited a high specific surface area of 60 m²·g⁻¹, leading to a maximum removal rate of MG up to 97.5%, and achieving an ultra-high adsorption capacity of 5850 mg·g⁻¹. Further kinetic study using the Langmuir isotherm and pseudo-second-order kinetic models confirmed that the adsorption process was a monolayer adsorption governed by chemisorption. This work offers an efficient and low-cost solution for dye wastewater treatment.

Keywords: Porous calcium carbonate, Adsorption, Malachite green, Water treatment

1. Introduction

Malachite Green (MG) is an inexpensive cationic dye widely used in textiles and dyeing industries. Due to its severe environmental toxicity and high carcinogenic risk, the removal of MG from dye wastewater is particularly important. To date, several methods for MG removal have been reported, including electrochemistry [1], photochemistry

[2], biodegradation [3], adsorption [4], and so on. Among them, the adsorption method is highly favored by researchers, owing to its advantages of high efficiency, low cost, and strong operability [5]. In recent years, numerous adsorbents have been developed, such as activated carbon [6], zeolites [7], clay [8], MOFs [9], and surfactant modified ferrite nanoparticle [10]. However, many of these adsorbents possess limited adsorption capacities, necessitating the development of more efficient adsorbents.

Herein, a porous CaCO_3 was synthesized via a facile template carbonization strategy with sodium dodecyl sulfate (SDS) as the template and CO_2 as the carbonization agent. The obtained porous CaCO_3 achieved an exceptional MG adsorption performance with an ultra-high adsorption capacity of up to $5850 \text{ mg}\cdot\text{g}^{-1}$. Different isothermal adsorption and kinetic models were employed to elucidate the adsorption mechanism. The high efficiency, low cost, and environmental compatibility of the porous CaCO_3 offer a promising solution for MG removal in dye wastewater treatment.

2. Experimental

Firstly, limestone, as the raw material, was calcined in a resistance furnace at 1000°C for 2 h to produce CaO . Then a digestive process was conducted under 60°C for 1 h using a mass ratio of 14:1 of deionized water to CaO . The obtained $\text{Ca}(\text{OH})_2$ slurry was filtered through a 200 mesh sieve to obtain a uniform dispersion. Subsequently, $0.1 \text{ mol}\cdot\text{L}^{-1}$ SDS solution was added into the $\text{Ca}(\text{OH})_2$ slurry as template agent with a molar ratio of 1:1. After standing for 10 min, CO_2 was injected continuously with a flow rate of $300 \text{ mL}\cdot\text{min}^{-1}$ under 30°C for carbonization until the pH value of the solution drops to 7. The obtained dispersion is filtered to collect the residue, which is washed three times to remove the template agent. Finally, the sample is dried to constant weight in an oven at 105°C to obtain the porous CaCO_3 . Details on the reagents, characterization methods, and adsorption tests are provided in the Supporting Information.

3. Results and discussion

The X-ray diffractometer (XRD) pattern of the obtained porous CaCO_3 (Fig 1a)

matches well with the standard card of PDF#05-0586, indicating a calcite phase. The SEM image (Fig 1b) shows that the porous CaCO_3 has a nanoparticle structure, which can provide abundant adsorption sites. The BET test results (Fig 1c and d) further reveal that the porous CaCO_3 possesses a high specific surface area of $60 \text{ m}^2 \cdot \text{g}^{-1}$, higher than most other works (Table S1). The pores in the porous CaCO_3 are mainly micro- to mesoporous, with an average pore size of 3.8 nm and a pore volume of $0.33 \text{ cm}^3 \cdot \text{g}^{-1}$. Due to the high surface area, it is expected to be an efficient adsorbent.

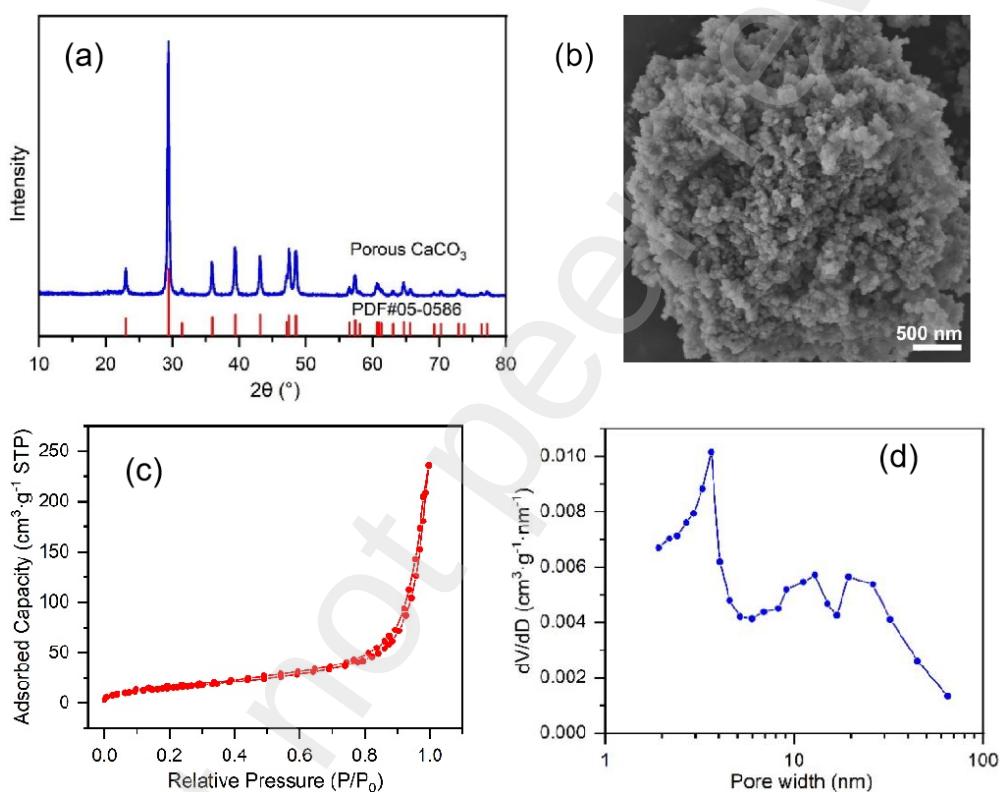


Fig 1. (a) XRD pattern, (b) SEM image, (c) N_2 adsorption/desorption isotherms, and (d) pore size distribution of the porous CaCO_3

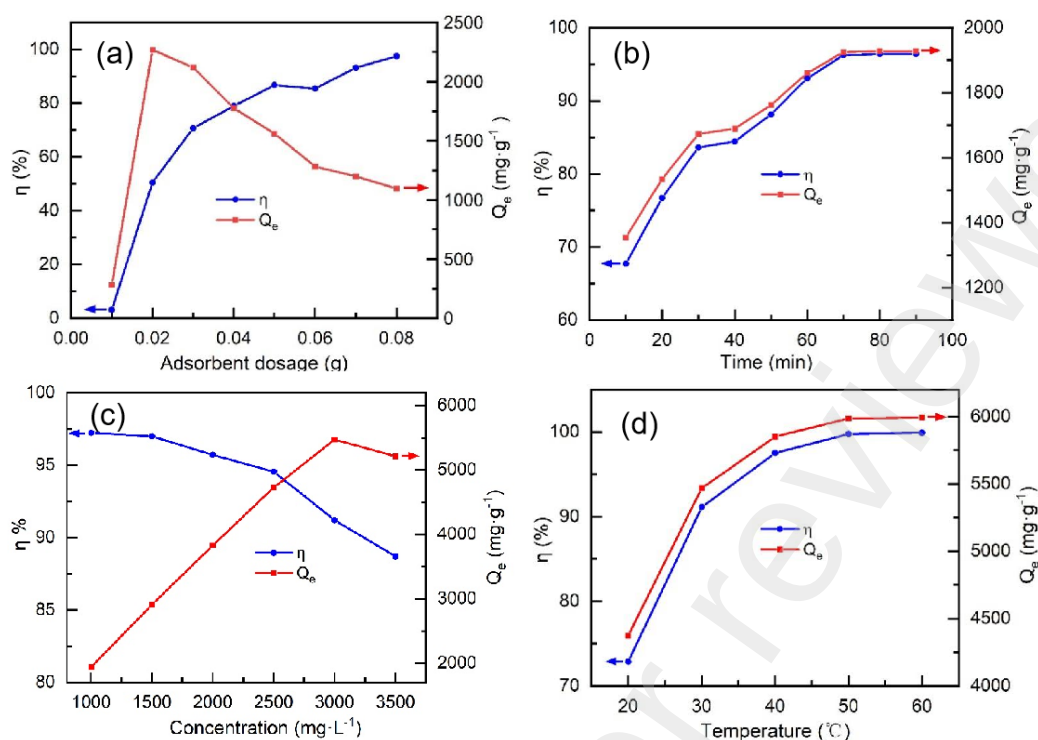


Fig 2. Effect of (a) adsorbent dosage, (b) adsorption time, (c) initial concentration, and (d) adsorption temperature on the adsorption performance of porous CaCO_3 for MG

The effect of various adsorption conditions, including the dosage of porous CaCO_3 , adsorption time, initial concentration of MG solution and adsorption temperature, on the adsorption performance is shown in Fig 2.

Fig 2a shows that the removal rate increases with the increasing of adsorbent dosage. Whereas, the adsorption capacity decreases with the increasing of adsorbent dosage. It suggests that 0.05 g of porous CaCO_3 can achieve a removal rate of 86.71% and an adsorption capacity of $1560.8 \text{ mg}\cdot\text{g}^{-1}$, without leading to excessive waste of adsorbent or making the removal rate too low.

For the effect of adsorption time, it is easy to understand that sufficient time will increase the removal rate and adsorption capacity until adsorption equilibrium is reached. As shown in Fig 2b, the adsorption process can be divided into three stages. Initially, in the first 30 min, the concentration of MG in the solution is much higher than that on the surface of porous CaCO_3 , generating a more powerful driving force of mass transfer, as well as providing sufficient active sites, leading to a rapid stage of MG removal. The second stage, from 30 to 70 min, is relatively slow, during which

the concentration of MG on the surface of porous CaCO_3 gradually approaches that in the solution, increasing the mass transfer resistance. It should be noted that some active sites are still not saturated during this stage, resulting in a slower adsorption process. The third stage, after 70 min of adsorption, is a dynamic equilibrium process, in which the removal rate and adsorption capacity remain stable at around 96 % and $1925 \text{ mg}\cdot\text{g}^{-1}$ respectively.

In Fig 2c, the removal rate of MG decreased with the increasing of the initial concentration. However, the absorption capacity increased gradually with an initial concentration no more than $3000 \text{ mg}\cdot\text{L}^{-1}$, showing a maximum adsorption capacity of $5472.7 \text{ mg}\cdot\text{g}^{-1}$ and a removal rate of 91.2%. This is owing to the fact that when the initial concentration increases, it means more MG needed to be removed, thus the calculated MG removal rate decreases naturally. In addition, the adsorbed MG on the surface of porous CaCO_3 is thought to have a repulsive effect on the MG in the solution, hindering the adsorption process [5].

Higher temperature usually maintained the molecules more active, promoting the adsorption process. In Fig 2d, the removal rate and absorption capacity are positively correlated with adsorption temperature, which are around 97.5% and $5850 \text{ mg}\cdot\text{g}^{-1}$ at 40°C respectively. It might be attributed to high surface area of porous CaCO_3 , revealing more active sites for adsorption of MG.

To investigate the adsorption mechanism, the adsorption isotherm and kinetic have been performed respectively. The adsorption equilibrium isotherm was fitted by with Langmuir (equation 1) and Freundlich (equation 2) models in Fig 3a and 3b, which are employed to express monolayer and multilayer adsorption process respectively [11]. The fitting parameters were listed in Table S2.

$$\frac{C_e}{Q_e} = \frac{1}{K_L Q_{max}} + \frac{C_e}{Q_{max}} \quad (1)$$

$$\ln Q_e = \ln K_F + \frac{\ln C_e}{n} \quad (2)$$

where C_e ($\text{mg}\cdot\text{L}^{-1}$) expresses the concentration of MG solution at equilibrium; Q_{max} and Q_e ($\text{mg}\cdot\text{g}^{-1}$) denote the maximum and equilibrium adsorption capacity of MG

respectively; K_L ($\text{L} \cdot \text{mg}^{-1}$) represents the Langmuir constant, correlated with the adsorption bonding energy; K_F ($\text{mg}^{1-1/n} \cdot \text{L}^{1/n} \cdot \text{g}^{-1}$) is the Freundlich constant; n is the Freundlich adsorption strength constant.

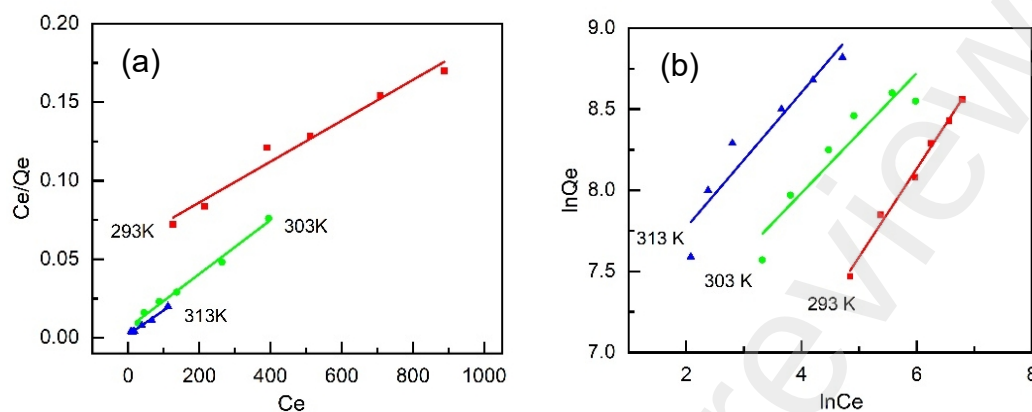


Fig. 3. Fitting lines of adsorption isotherm of MG by porous CaCO_3 . (a) Langmuir model, (b) Freundlich model

It can be found that the adsorption constants, both K_L and K_F , increase with the increasing of temperature. In other words, higher temperature is more beneficial to the adsorption reaction [12], which met with the results of effect of adsorption temperature in Fig 2d well. Both of Langmuir and Freundlich isotherm model showed a good linear correlation with high level of coefficient (R^2), indicated the adsorption of MG by porous CaCO_3 might be a complicated monolayer and multilayer adsorption process. It is worth noting that the fitting coefficient of Langmuir isotherm model is significantly higher than Freundlich model at high temperature, suggesting the adsorption of MG by porous CaCO_3 is mainly monolayer adsorption.

The adsorption kinetics of is employed to analyze the relationship between adsorption capacity and adsorption time during the adsorption process. Herein, the pseudo-first-order (equation 3) and pseudo-second-order (equation 4) [13] kinetic equations are used to fit the experimental results (Fig. 4), from which the equilibrium adsorption capacity of MG can be predicted. And the fitting parameters were listed in Table S3.

$$\ln (Q_e - Q_t) = \ln Q_e - k_1 t \quad (3)$$

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \quad (4)$$

where Q_e and Q_t ($\text{mg}\cdot\text{g}^{-1}$) denote the adsorption capacity of MG at equilibrium and any contact time t (min); k_1 (min^{-1}) and k_2 ($\text{g}\cdot\text{mg}^{-1}\cdot\text{min}^{-1}$) are the equilibrium rate constant of pseudo-first-order and pseudo-second-order kinetic, respectively.

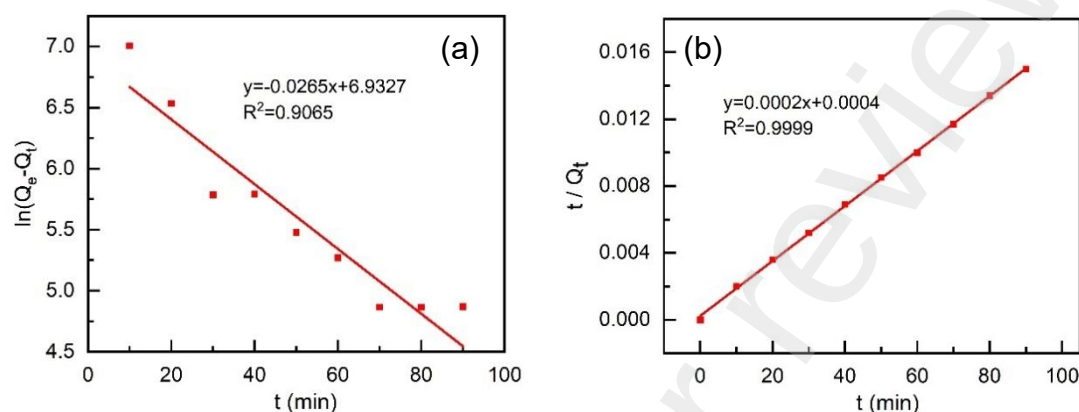


Fig. 4. Fitting curves of adsorption kinetic of MG on porous CaCO_3 by pseudo-first-order (a) and pseudo-second-order (b) adsorption kinetic model

The fitting results showed that the experimental data have good linear correlation with the coefficients (R^2) higher than 0.9, both pseudo-first-order and pseudo-second-order kinetic model. To be precise, the R^2 of pseudo-second-order model is close to 1, higher than that of the pseudo-second-order kinetics model, suggested the adsorption of MG on porous CaCO_3 is mainly controlled by chemisorption [14]. The XRD and FTIR of the porous CaCO_3 after adsorption with MG were characterized in Fig S1 and S2, confirming the MG has been adsorbed onto the porous CaCO_3 successfully.

4. Conclusions

Porous CaCO_3 with a high specific surface area of up to $60 \text{ m}^2\cdot\text{g}^{-1}$ has been prepared via a facile template carbonization strategy. It shows excellent chemisorption capability for MG, achieving a maximum removal rate of 97.5% and an ultra-high saturation adsorption capacity of $5850 \text{ mg}\cdot\text{g}^{-1}$, respectively. The kinetics study indicates that this adsorption process conforms to a monolayer chemical adsorption mechanism. This work provides a simple approach for the development of

efficient porous calcium carbonate adsorbents.

Acknowledgments

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